Chenming Tang

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Research Interests

#Memory #RAG #Personalization #NLP Applications #Agent

Education

Peking University Beijing, China
M.S. (Academic) in Computer Science, GPA: 3.60/4.00 *2024–2027*

Peking University Beijing, China
B.S. in Computer Science, GPA: 3.72/4.00 *2020–2024*

Publications

Refereed Publications

1. **Chenming Tang**, Hsiu-Yuan Huang, Weijie Liu, Clive Bai, Saiyong Yang, Yunfang Wu. "Do Not Step Into the Same River Twice: Learning to Reason from Trial and Error". *ACL 2026 (Oral)*.
2. **Chenming Tang***, Yutong Yang*, Kexue Wang, Yunfang Wu. "Aligning Language Models with Real-time Knowledge Editing". *ACL 2026*.
3. Hsiu-Yuan Huang*, **Chenming Tang***, Weijie Liu, Clive Bai, Saiyong Yang, Yunfang Wu. "Think Outside the Policy: In-Context Steered Policy Optimization". *ACL 2026 (Findings)*.
4. **Chenming Tang***, Zhixiang Wang*, Hao Sun, Yunfang Wu. "Large Language Models Might Not Care What You Are Saying: Prompt Format Beats Descriptions". *EMNLP 2025 (Findings)*.
5. **Chenming Tang**, Zhixiang Wang, Yunfang Wu. "SCOI: Syntax-augmented Coverage-based In-context Example Selection for Machine Translation". *EMNLP 2024*.
6. **Chenming Tang**, Fanyi Qu, Yunfang Wu. "Ungrammatical-syntax-based In-context Example Selection for Grammatical Error Correction". *NAACL 2024*.
7. **Chenming Tang**, Xiuyu Wu, Yunfang Wu. "Are Pre-trained Language Models Useful for Model Ensemble in Chinese Grammatical Error Correction?". *ACL 2023 (Short Paper)*.

Preprints

1. **Chenming Tang**, Jiawei Han. "Select-And-Extract: A Lightweight Plugin for Retrieval-Augmented Generation". *arXiv:2608.xxxxx*.
2. **Chenming Tang**, Hsiu-Yuan Huang, Weijie Liu, Junqiang Zheng, Saiyong Yang, Yunfang Wu. "Democratizing Tool Learning with Environments Fully Simulated by a Free 8B Language Model". *arXiv:2604.17739*.
3. Kun Liang, **Chenming Tang**, Clive Bai, Weijie Liu, Saiyong Yang, Yunfang Wu. "ADWIN: Adaptive Windows for Horizon-Aware On-Policy Distillation". *arXiv:2605.28396*.
4. Hsiu-Yuan Huang, Weijie Liu, **Chenming Tang**, Sanwoo Lee, Kai Yang, Yangkun Chen, Saiyong Yang, Yunfang Wu. "RLVR Datasets and Where to Find Them: Tracing Data Lineage for Better Training Data". *arXiv:2605.26971*.
5. Kun Liang, Clive Bai, Xin Xu, **Chenming Tang**, Sanwoo Lee, Weijie Liu, Saiyong Yang, Yunfang Wu. "ORBIT: On-policy Exploration-Exploitation for Controllable Multi-Budget Reasoning". *arXiv:2601.08310*.
6. Hao Sun, **Chenming Tang**, Gengyang Li, Yunfang Wu. "Lost in the Passage: Passage-level In-context Learning Does Not Necessarily Need a 'Passage'". *arXiv:2502.10634*.
7. Fanyi Qu, **Chenming Tang**, Yunfang Wu. "Evaluating the Capability of Large-scale Language Models on Chinese Grammatical Error Correction Task". *arXiv:2307.03972*.

Research Experience

University of Illinois Urbana-Champaign *Mar 2026 – Jul 2026*
Advisor: Prof. Jiawei Han Remote

- **Agent Memory:** Explored approaches for agent memory inspired by data stream mining and relational database; identified limitations in non-factual memory and repositioned to retrieval-augmented generation.
- **Retrieval-augmented Generation:** Proposed a simple, lightweight, training-free, yet effective plugin for RAG that directly addresses the two fundamental failures of RAG and overcomes the limitations of prior approaches.

Tencent & Peking University *Jul 2025 – May 2026*
Advisor: Dr. Weijie Liu & Prof. Yunfang Wu Beijing, China

- **Agentic Reinforcement Learning:** Democratized RL of tool-calling agents by adaptive environments fully simulated by open-source LMs, not requiring any annotated data, synthesized environments, or proprietary LMs.
- **Reasoning-oriented Reinforcement Fine-tuning:** Designed LTE, a mistake-hinting mechanism that helps the policy model solve more difficult problems during rollouts, significantly boosting the reasoning performance of post-trained models (ACL 2026 Oral).

Peking University *Mar 2022 – Jul 2025*
Advisor: Prof. Yunfang Wu Beijing, China

- **Knowledge Editing:** Led the development of CRAFT, a dynamic dataset for real-time and real-world knowledge editing, and proposed KEDAS, a novel self-adaptive paradigm of aligning LMs with real-time and real-world knowledge editing that significantly outperforms existing approaches (ACL 2026).
- **Prompting and In-context Learning:** Improved LMs' GEC performance with in-context demonstrations selected based on ungrammatical syntactic similarity (NAACL 2024), enhanced LMs on machine translation by in-context examples with a high coverage in both syntax and words (EMNLP 2024), and demonstrated a proper prompt format without well-designed descriptions is effective enough to achieve promising performance on many tasks (EMNLP 2025 Findings).
- **Grammatical Error Correction:** Highlighted that LM perplexity does not necessary correlate with grammatical correctness (ACL 2023 Short Paper).

Awards

- **Outstanding Graduate**, Peking University *2024*
- **Exceptional Award for Academic Innovation**, Peking University *2023*
- **Merit Student**, Peking University *2023*
- **First prize**, NLPCC 2023 Shared Task 1 (Chinese Grammatical Error Correction) *2023*
- **Second place**, CCL 2022 Shared Task 4 (Multi-reference Multi-source Chinese Learner Text Correction) *2022*

Teaching

Teaching Assistant | Introduction to Computing (C) *Spring 2025*
Peking University | Instructor: Prof. Zhifang Sui & Prof. Yunfang Wu Beijing, China

Standardized Tests

- **TOEFL iBT:** 111 (R:30, L:28, S:23, W:30) *Dec 2025*
- **GRE General Test:** 336 (V:166, Q:170, AWA:3.5) *Jul 2025*